# Impedance-Based Localization of Injected Attack ECUs in In-Vehicle CAN Networks*

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**Abstract**

The Controller Area Network (CAN) bus is widely used for real-time communication among in-vehicle Electronic Control Units (ECUs). Since CAN_H and CAN_L form a differential pair and multiple ECUs are connected through stubs, the CAN bus can be interpreted not only as a communication network but also as a physical transmission-line structure. From this perspective, each ECU transceiver and its internal circuit act as an impedance load on the bus. When an unauthorized attack ECU is additionally inserted into the CAN network, the existing branch configuration and impedance distribution of the bus may change, which can lead to signal reflection, attenuation, and waveform distortion. This work investigates a passive impedance-based localization concept for estimating the physical connection distance or location of an additionally inserted attack ECU in an in-vehicle CAN network. The proposed approach focuses on the additional ECU insertion scenario, where legitimate ECUs remain attached to the CAN bus while an unauthorized ECU is newly connected. In this scenario, the inserted ECU introduces a new branch and electrical load into the existing bus structure. These location-dependent physical-layer changes can be reflected in ordinary CAN communication signals and may provide useful clues for identifying where the unauthorized ECU is connected. Prior work has shown that physical-layer changes in CAN networks can be exploited for device detection and localization. Previous studies have shown that physical-layer changes in CAN networks can be exploited for device detection and localization. One prior work uses active Time Domain Reflectometry (TDR) to detect unknown CAN devices and estimate their cable distances, while another work uses deep features extracted from voltage signals to detect physical intrusions and localize their connection points on an in-vehicle CAN bus. Unlike active TDR-based approaches, which inject a separate probing pulse and estimate distance from reflection delay, the proposed concept aims to use

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ordinary CAN communication signals without additional signal injection. The CAN bus is modeled as a distributed-parameter transmission line, and location-dependent waveform changes are interpreted as clues caused by impedance mismatch, reflection, and attenuation. Candidate measurement features include CAN signal overshoot, rising time, falling time, ringing, edge distortion, and frequency-dependent impedance variation according to branch length and insertion position. Future work will implement a CAN bus testbed, insert an unauthorized ECU at controlled positions and branch lengths, and compare measured waveform features with transmission-line-model-based profiles and normal CAN bus baselines. The final goal is to evaluate whether passive localization of an additionally inserted attack ECU is feasible using only ordinary CAN communication signals, thereby supporting the identification and removal of unauthorized ECUs from in-vehicle CAN networks.

## Keywords

CAN, Impedance, Network Security, voltage, Radar

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